

Multiple Choice

1. **Answer:** A

Options: A) violet; B) turquoise; C) green; D) blue; E) red

Note: Correct option: A - violet

2. **Answer:** D

Options: A) fluorescence.; B) absorption.; C) resonance.; D) de-excitation.; E) refraction.

Note: Correct option: D - de-excitation.

3. **Answer:** A

Options: A) air resistance is always zero in free fall.; B) volume of both rocks is the same.; C) force due to gravity is the same on each.; D) gravitational constant is the same for both rocks.; E) force due to gravity is zero in free fall.

Note: Correct option: A - air resistance is always zero in free fall.

4. **Answer:** A

Options: A) greatly increases; B) slightly increases; C) is halved; D) becomes zero; E) greatly decreases

Note: Correct option: A - greatly increases

5. **Answer:** A

Options: A) -11.00 diopters; B) -9.50 diopters; C) -10.50 diopters; D) -9.00 diopters; E) -7.93 diopters

Note: Correct option: A - -11.00 diopters

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '15.9'.

7. **Answer:** False

Options: A) True; B) False

Note: Actual answer is ' $6.28 \cdot 10^{34} \text{ m}^{-3} \text{ s}^{-1}$ '.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'an electric field'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 240 cm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** energy of alpha particles.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key